

House Price Prediction

Predicting California housing prices with Linear Regression

Dataset: California Housing (scikit-learn)

Project Overview

A regression workflow from raw data to evaluated model

1

Load Data

Fetch the California Housing dataset and build a DataFrame

2

Explore

Check structure, summary statistics and missing values

3

Visualize

Correlation heatmap and feature distributions

4

Train

Split data and fit a Linear Regression model

5

Evaluate

MAE, RMSE, R^2 and residual analysis

Dataset

California Housing dataset — 20,640 rows, 9 columns

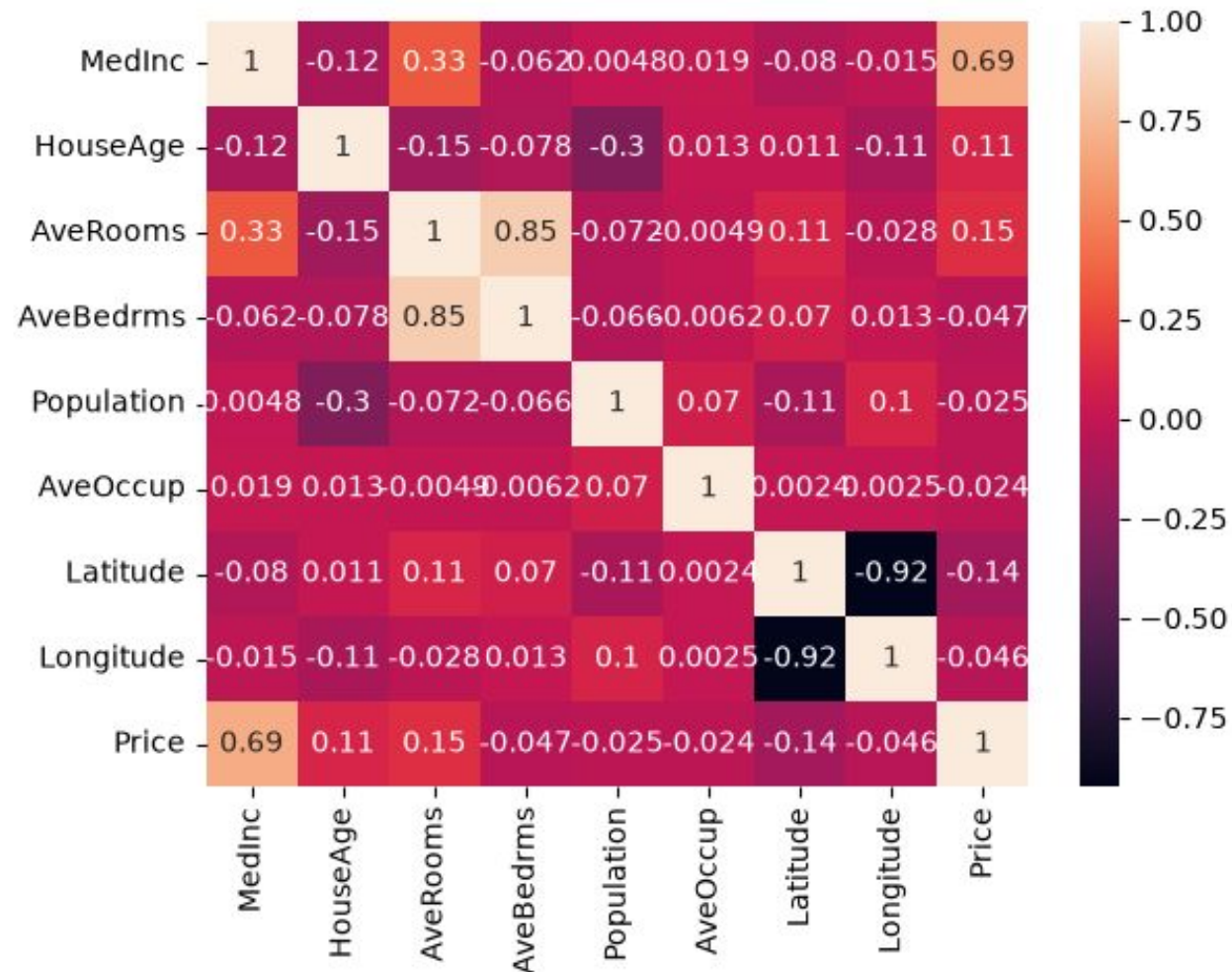
Each row represents a California census block group. Features describe income, housing age, rooms, population, occupancy and location; the target is median house price (in \$100,000s).

Feature	Mean	Std Dev	Min	Max
MedInc (median income)	3.87	1.90	0.50	15.00
HouseAge	28.64	12.59	1.00	52.00
AveRooms	5.43	2.47	0.85	141.91
AveBedrms	1.10	0.47	0.33	34.07
Population	1425.48	1132.46	3.00	35682.00
AveOccup	3.07	10.39	0.69	1243.33
Price (target)	2.07	1.15	0.15	5.00

No missing values were found in any column.

Feature Correlation

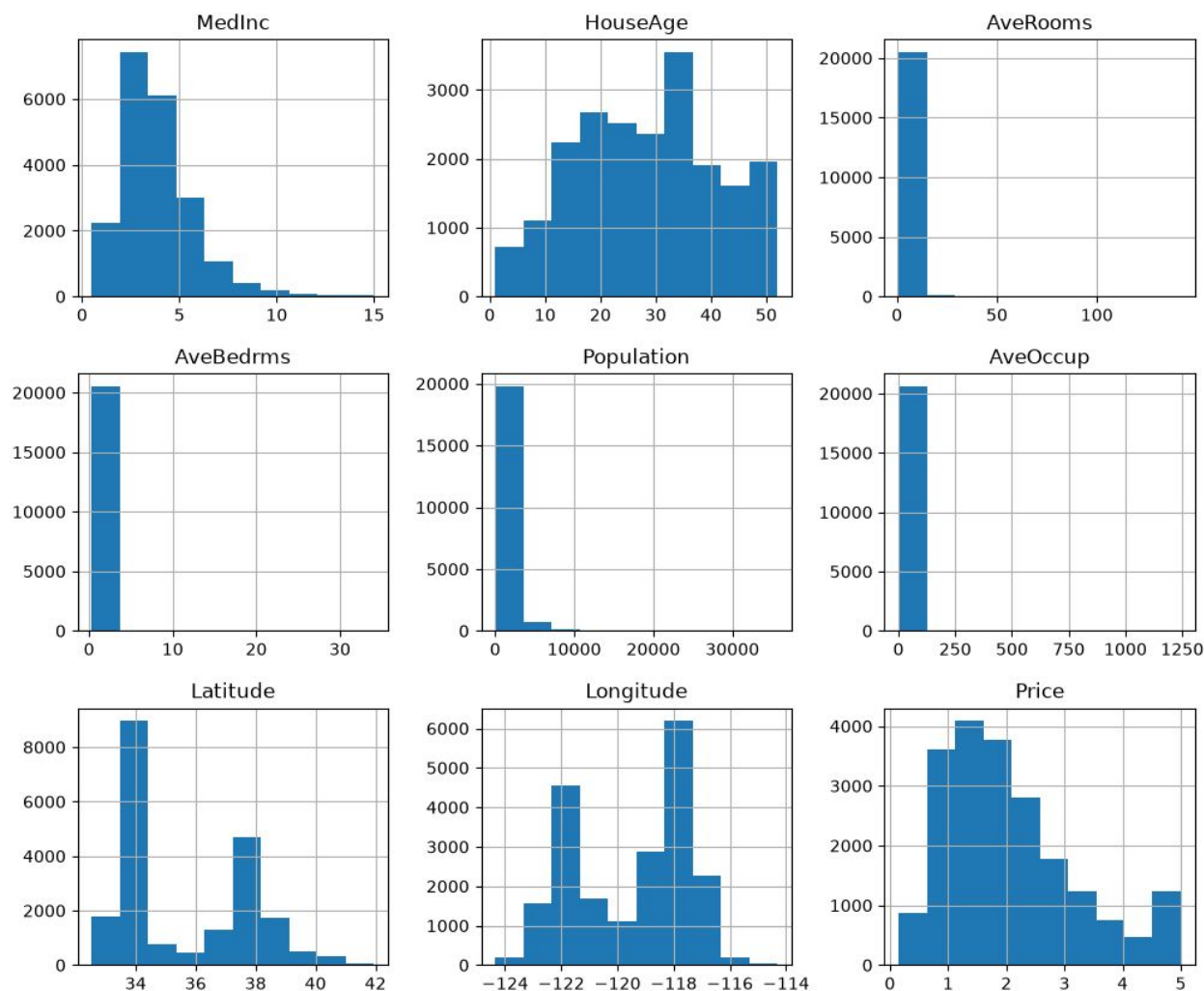
Correlation heatmap across all numeric features



Median income (MedInc) shows the strongest positive correlation with house price.

Feature Distributions

Histograms of all numeric variables



Model Setup

Preparing data and training the model

Features (X)

8 columns: MedInc, HouseAge, AveRooms, AveBedrms, Population, AveOccup, Latitude, Longitude

Target (y)

Price — median house value per block group

Train / Test Split

80% training, 20% testing (random_state = 42)

Model

Linear Regression (scikit-learn)

Model Evaluation

Performance on the held-out test set

0.533

MAE

Mean Absolute Error

0.746

RMSE

Root Mean Squared Error

0.576

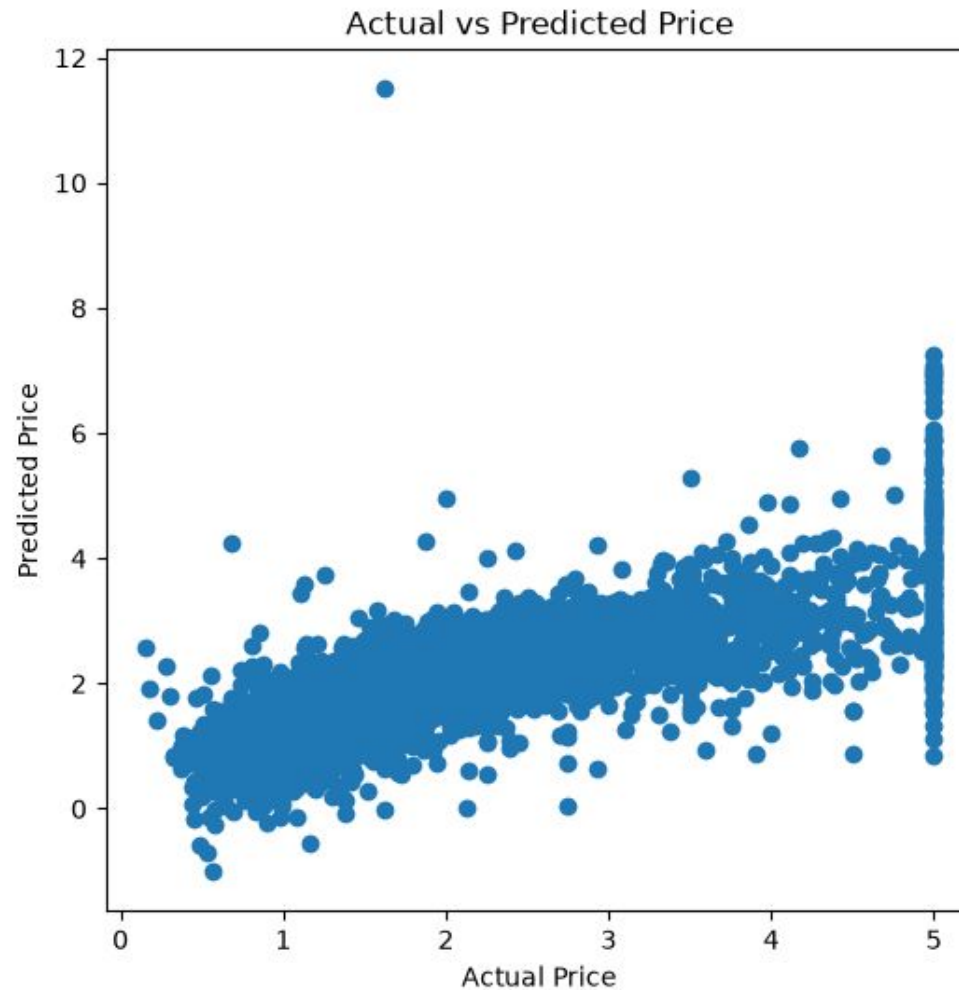
R²

Variance explained by the model

The model explains about 58% of the variance in house prices on unseen data.

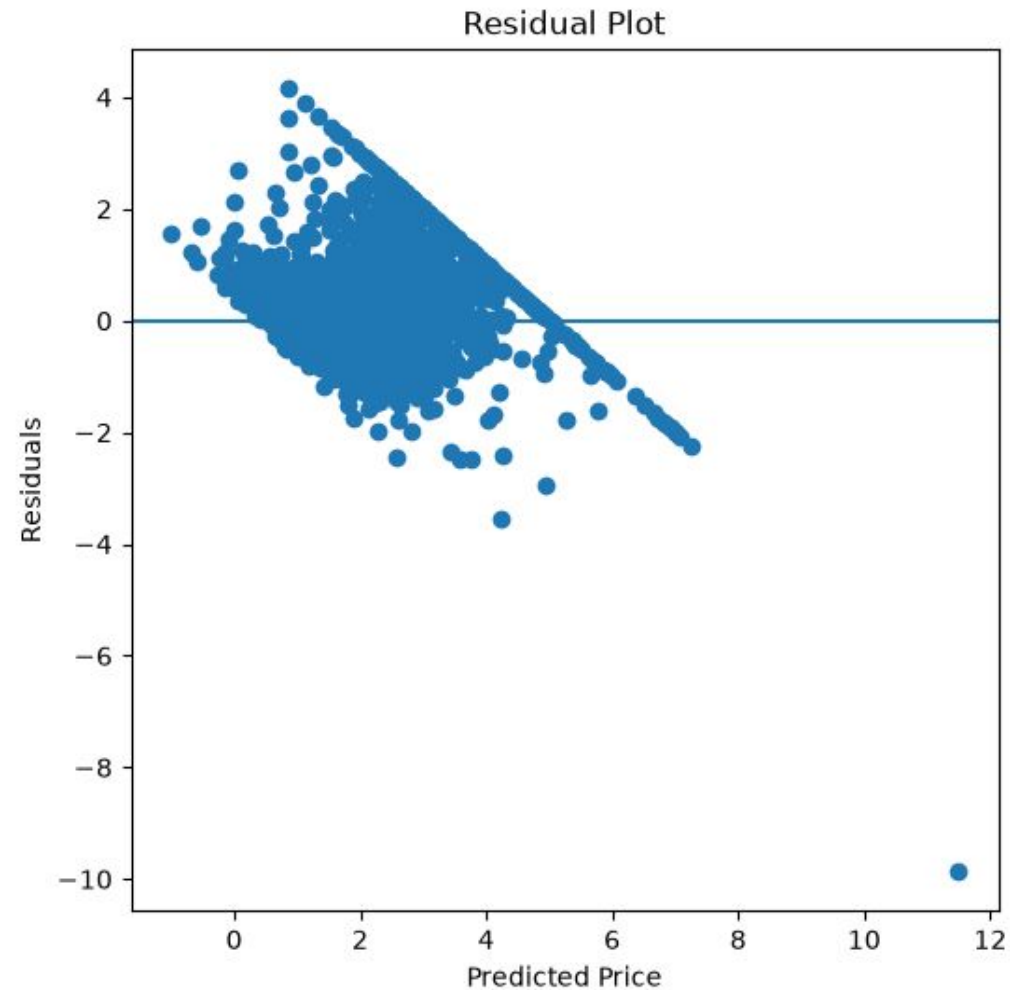
Actual vs. Predicted Prices

How closely predictions track true values



Residual Analysis

Prediction errors plotted against predicted values



Residuals scatter fairly evenly around zero, with a wider spread at higher predicted prices.

Summary

- A Linear Regression model was trained on the California Housing dataset (20,640 records, 8 features).
- Median income is the feature most strongly correlated with house price.
- On the test set, the model achieved an R^2 of 0.576, MAE of 0.533 and RMSE of 0.746.
- Residuals are reasonably centered around zero, though variance increases for higher-priced homes, suggesting a non-linear model could improve accuracy further.